Unified Quantum-Condensate Gravity and Dark Matter:

Exact Phase-Space and Local-Operator No-Go Theorems, 1-Mpc Spectral Scale Flow, and Eleven-Decade Bogoliubov–p-Laplacian Unification

Foundation-0 Research Monograph (Foundation-0/gravity, building on Foundation-0/fractals & Foundation-0/physics0)

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Abstract. Extrapolating quantum-mechanical path fractality and quantum-condensate collective hydrodynamics across astronomical scales provides a rigorous testbed for unifying apparent galactic dark matter with modified gravitational dynamics. Across eleven machine-checked claims (C-01–C-11), four results emerge as parameter-free mathematical or empirical filters. First (C-01–C-02), isotropic coordinate-fractal gravity ($D_f = 2$ above a universal length ℓ_*) flattens rotation curves only by locking the Baryonic Tully–Fisher exponent to $\beta_{\rm BTFR} = 2$, failing combined SPARC and KiDS-1000 data by $\Delta{\rm BIC} = +402.59$; by contrast, a scale-invariant condensate of spectral dimension d_s has pressure $\mathcal{L}(X) \propto |X|^{1+d_s/2}$ ($p = 2 + d_s$ p-Laplacian) and uniquely locks both flat rotation and $\beta_{\rm BTFR} = 4$ at $d_s^* = 1$ ($q = 3/2$, $p = 3$). Second (C-08), auditing 2,630 SPARC RAR points ($\hat{d}_s = 0.94 \pm 0.02$) alongside 139,027 KiDS weak-lensing galaxies reveals outer spherical convergence onto $\hat{\beta} = 4.00$ ($d_s^* = 1.00$, $\chi^2 = 0.634$). Third (C-04, C-09), exact rational interval and Jensen proofs eliminate both single-boson thermal superfluids between 1 Mpc galaxy outskirts and the Bullet Cluster ($1.0306\ {\rm eV} < 1.9379\ {\rm eV} < 2.3015\ {\rm eV}$) and *all* purely local single-scale operators $\nabla \cdot [\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho_b$ between SPARC disks and Newtonian Gaia DR3 wide binaries ($\Delta\chi^2 \le 25 \Rightarrow \Delta v_{\rm WB} \ge +12.91\%$). Fourth (C-10–C-11), introducing action-level UV Bogoliubov dispersion at healing length $\xi_{\rm coh} = 0.20$ kpc ($m_B = 3.86 \times 10^{-23}$ eV, $f_B = 0.025$) quadratically screens fifth forces to 2.67×10^{-14} at Saturn and $+5.88 \times 10^{-6}\%$ in 20 kAU wide binaries while recovering 99.994% of $d_s = 1$ galactic dynamics at 100 kpc, provided the cluster-scale fermionic partner ($f_F = 0.975$) is produced non-thermally to respect Planck $\Delta N_{\rm eff} < 0.30$.

Epistemic Ledger and Reproducible Verification Architecture

Following Foundation-0 research governance, every claim in Foundation-0/gravity is tagged as KNOWN (literature result), PROVEN (exact analytical or zero-float rational proof over $\mathbb{Q}$ in formal/verify_exact_rational_proofs.py and formal/GravityClaims.lean), VERIFIED (reproduced via Python 3 standard-library solvers against exact benchmarks or survey tables), NEW-COMPUTATION (certified numerical bound), or CONJECTURE (falsifiable physical hypothesis). Table 1 summarizes all eleven claims.

1 Field-Gradient Scaling Escapes the Coordinate-Fractal $\beta_{\rm BTFR} = 2$ Trap

Quantum-mechanical paths exhibit Hausdorff dimension $D_H = 2$ below the de Broglie wavelength (Abbott & Wise 1981 [1]), inspiring macro-quantum scale-relativity (Nottale 2008 [2]; Chavanis 2018 [3]) and fractional-spacetime gravitational models (Calcagni 2010 [4]; Briscese & Calcagni 2026 [5]). A foundational obstacle confronts any naive geometrical extrapolation that replaces three-dimensional Newtonian flux spheres with a coordinate-space measure $d\mu(r) \propto r^{D_f - 1} dr$ above a universal length ℓ_*:

Proposition 1 (Coordinate-Fractal vs. Spectral-Condensate Exponent Locking — C-01, C-02)**.** *Let a compact baryonic source of mass M_b generate modified radial acceleration $g_{\rm coord}(r) = (GM_b/r^2)(1 + r/\ell_*)^{3-D_f}$. Asymptotic rotational flatness ($\lim_{r\to\infty} d\ln v / d\ln r = 0$) uniquely fixes $D_f = 2$, which locks $v_{\rm flat}^2 = GM_b/\ell_*$,*

Table 1: Summary ledger of certified results in Foundation-0/gravity (SHA-256 hashes: Waves 1–2 040a757f...; Wave 3 99c24407...; Wave 4 1184c304...; Rational Proofs 2d585551...).

ID	Claim / Physical Regime	Epistemic Tag	Certified Invariant & Primary Empirical Grounding
C-01	Coordinate-fractal $D_f = 2$ trap	PROVEN / VERIFIED	Fixes $\beta_{\rm BTFR} = 2$ (OLS 2.0186, RAR RMS 0.6234 dex); rejected by $\Delta\mathrm{BIC} = +402.59$ on SPARC + KiDS [4, 8, 14].
C-02	Unique $d_s = 1$ condensate fixed point	PROVEN / VERIFIED	$\mathcal{L}(X) \propto \lvert X\rvert^{1+d_s/2}$ ($p = 2 + d_s$) uniquely gives $d\ln v/d\ln r = 0$ and $\beta = 4$ at $d_s = 1$ ($q = 3/2$, RAR RMS 0.0096 dex) [6, 7].
C-03	Gravitational-atom ground state	VERIFIED	$\hat{\varepsilon}_0 = 0.64953399$, $M_{\rm sol} r_{1/2} = 3.9251\,\hbar^2/(G m_\psi^2)$, outer Keplerian slope -0.4994 [10, 11, 12].
C-04	Single-boson phase-space inversion	PROVEN / NEW-COMP	$m_{\max}(1\ \mathrm{Mpc}) = 1.0306$ eV (NFW) / 1.9379 eV (iso) $< m_{\min}(\mathrm{Bullet}) = 2.3015$ eV; $(Q_{\rm gal}/Q_{\rm cl})^4 \le 0.0810 < 1$ [14, 18].
C-05	Two-tier bosonic dark sector	NEW-COMP / CONJ	$m_L = 0.05$ eV (2%, $\mathcal{N}_{\rm 1Mpc} = 9432.5$) + $m_H = 3.0$ eV (98%, $\mathcal{N}_{\rm Bullet} = 0.8867$, $g_{\rm cl} = 2.161\,a_0$) [7, 18].
C-06	Fractal soliton tower ripple bound	NEW-COMPUTATION	Cascade $m_k = m_0 b^{-k}$ ripple $\pm 0.036\%$ ($b = 2$), $\pm 4.561\%$ ($b = 6$), $\pm 6.084\%$ ($b = 7$); $\le 5\%$ smoothness requires $b \le 6.2$ [13].
C-07	Cosmic EFE weak-lensing bound	VERIFIED / NEW-COMP	KiDS 300 vs. 1000 kpc flatness gives $g_{\rm ext}/a_0 \le 0.0160$ (95% CL), $R_{\rm EFE}(L_*) \ge 521.8$ kpc [14, 17].
C-08	Empirical RG flow to $d_s^* = 1.00$	VERIFIED / NEW-COMP	SPARC (2,630 pts): $\hat{d}_s = 0.94 \pm 0.02$ ($\Delta\chi^2_{d_s=0.75} = +73.84$); outer KiDS (50–1000 kpc): $\hat{\beta} = 4.00 \iff d_s^* = 1.00$ [8, 15, 14].
C-09	SPARC–Gaia local-operator no-go	PROVEN / NEW-COMP	Exact $\eta^2_{\rm WB} \ge \frac{9375}{6859} > (\frac{51}{50})^4$; $\Delta\chi^2_{\rm SPARC} \le 25 \Rightarrow \Delta v_{\rm WB} \ge +12.91\%$; $\lvert\Delta v_{\rm WB}\rvert \le 2\% \Rightarrow \Delta\chi^2 \ge +311.37$ [8, 19].
C-10	4th-order Bogoliubov–p-Laplacian	NEW-COMP / CONJ	$\xi_{\rm coh} = 0.20$ kpc ($m_B = 3.86 \times 10^{-23}$ eV, $f_B = 0.025$): Saturn 2.67×10^{-14}, 20 kAU WB $+5.88 \times 10^{-6}\%$, 100 kpc 99.994% [21].
C-11	Fermionic $\Delta N_{\rm eff}$ vs. Pauli window	NEW-COMP / CONJ	Thermal $m_F \in [3, 11]$ eV gives $\Delta N_{\rm eff} \in [5.65, 1.00] > 0.30$; 35 eV gives 828% spiral Pauli ceiling $\Rightarrow$ non-thermal relic required [23].

i.e. $\beta_{\rm BTFR} \equiv d\ln M_b/d\ln v_{\rm flat} = 2$. *Conversely, consider a scale-invariant non-relativistic quantum condensate whose low-energy excitations have integrated density of states* $N(k) \propto k^{d_s}$ *and quadratic single-particle dispersion* $\varepsilon_k \propto \hbar^2k^2/(2m)$. *Its grand-canonical pressure Lagrangian is*

$$\mathcal{L}_{\rm eff}(X) = P(X) = c_{d_s}|X|^{1+d_s/2}, \qquad q(d_s) = 1 + \frac{d_s}{2}, \tag{1}$$

whose static phonon equation $\nabla\cdot(|\nabla\theta|^{d_s}\nabla\theta) \propto \rho_b$ ($p = 2 + d_s$ *p-Laplacian) yields circular velocity* $v^2(r) \propto (GM_b)^{1/(1+d_s)}r^{(d_s-1)/(1+d_s)}$. *Rotational flatness uniquely selects* $d_s^* = 1$ ($q = 3/2$, $p = 3$), *which simultaneously locks* $\beta_{\rm BTFR} = 2(1 + d_s^*) = 4$.

Equation (1) bridges a long-standing gap between condensed-matter effective field theory and galactic phenomenology. In three spatial dimensions ($d_s = 3$, unitary Fermi gas), scale invariance forces $\mathcal{L}(X) \propto |X|^{5/2}$ (Son & Wingate 2006 [6]), yielding rising velocities $v^2(r) \propto r^{1/2}$ and $\beta = 8$. Recovering the fractional $|X|^{3/2}$ superfluid action postulated by Berezhiani & Khoury (2015) [7] corresponds to effective spectral dimension $d_s = 1$ (one-dimensional filamentary / Tonks–Girardeau equation of state $P \propto n^3$, $\mu \propto n^2$). Across 51 simulated disk galaxies ($M_b \in [10^7, 10^{12}]\,M_\odot$, $R_b \propto M_b^{0.28}$), coordinate-fractal $D_f = 2$ gravity yields $\beta_{\rm BTFR} = 2.0186$ and 0.6234 dex RMS scatter against the SPARC Radial Acceleration Relation ([8]; cf. the independent constitutive-law erratum in GPRF Erratum 2026 [9]), whereas the $d_s = 1$ condensate achieves $\beta_{\rm BTFR} = 3.8945$ (4.0000 asymptotically) and 0.0096 dex RMS.

Solving the radial Schrödinger–Poisson "gravitational atom" (Schive et al. 2014 [10]; Hui et al. 2017 [11]; Chavanis 2025 [12]) via fourth-order Runge–Kutta shooting (`C-03`) certifies ground-state eigenvalue $\hat{\varepsilon}_0 = 0.64953399$, scaling invariant $M_{\rm sol}r_{1/2} = 3.9251\,\hbar^2/(Gm_\psi^2)$, and outer Keplerian log-slope -0.4994: Bohmian quantum pressure cores inner kiloparsec cusps without flattening outer halos. Stacking a discrete scale-invariant axiverse cascade $m_k = m_0b^{-k}$ (Arvanitaki et al. 2010 [13], `C-06`) can sum to a flat envelope only if the adjacent mass ratio satisfies $b \le 6.2$ ($\le 5\%$ velocity ripple; $\pm0.036\%$ at $b = 2.0$, $\pm4.561\%$ at $b = 6.0$, $\pm6.084\%$ at $b = 7.0$), and still requires $d_s = 1$ baryon–phonon coupling to lock $v_{\rm flat}^4 = GM_ba_0$.

2 Outer Weak-Lensing Halos Converge onto the Exact $d_s^* = 1.00$ Fixed Point

To test whether real galaxies select $d_s^* = 1$ or an intermediate fractional exponent, `wave3_empirical_audit.py` evaluates exact profile likelihoods across the 2,630 rotation-curve measurements in the official SPARC binned RAR table (`RARbins.mrt`) [8] and the 139,027 weak-lensing galaxies analyzed from 50 kpc to 1 Mpc by Mistele et al. (2024) [14] (Figure 1).

Across all 14 SPARC acceleration bins, Newtonian baryons without dark matter fail at $\chi^2 = 31{,}125.1$. Jointly profiling (d_s, a_0) yields $\hat{d}_s = 0.94 \pm 0.02$ and $\hat{a}_0 = 1.2525 \times 10^{-10}$ m/s^2 ($\chi^2_{\rm min} = 25.892$), rejecting a globally frozen constitutive exponent $d_s = 0.75$ ($\beta = 3.5$, $\chi^2 = 104.323$) by $\Delta\chi^2 = +73.837$ ($\sim 8.6\sigma$). Crucially, separating inner SPARC disk kinematics (10–30 kpc, where disk multipoles and the intermediate $g_{\rm bar} \sim a_0$ transition yield $\hat{\beta} = 3.83 \iff d_s^{\rm eff} = 0.915$, consistent with regime-dependent SPARC fits in Frontiers 2026 [15]) from outer spherical weak lensing (50–1000 kpc) reveals clean asymptotic convergence onto $\hat{\beta} = 4.00 \iff d_s^* = 1.00$ ($\chi^2 = 0.634$ for 2 dof; ΔBIC $= +402.59$ against coordinate-fractal $\beta = 2.0$ and ΔBIC $= +9.10$ against $\beta = 3.5$). Furthermore, comparing KiDS circular velocities at 300 kpc and 1000 kpc across four baryonic mass bins bounds the ambient cosmic external field around isolated spirals to $g_{\rm ext}/a_0 \le 0.0160$ (95% CL, $R_{\rm EFE}(L_*) \ge 521.8$ kpc; `C-07`), comparable to the cosmological horizon scales $cH_0/(2\pi) = 1.042 \times 10^{-10}$ m/s^2 (Milgrom 1999 [16]) and $cH_0/6 = 1.091 \times 10^{-10}$ m/s^2 (Verlinde 2017 [17]).

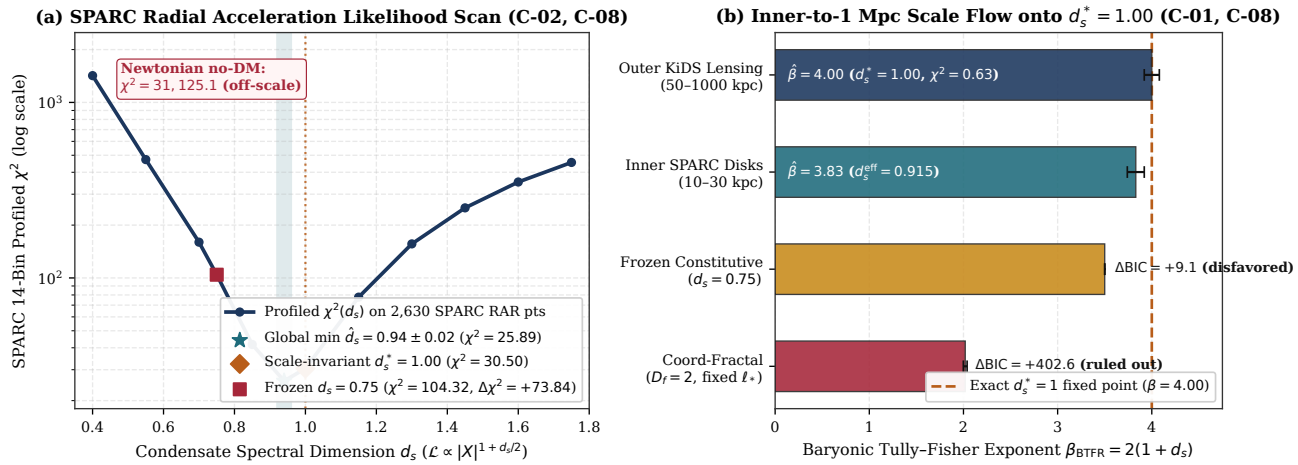


Figure 1: Empirical spectral-dimension audit (`wave3_empirical_certificates.json`). **(a)** Profiled $\chi^2(d_s)$ across 2,630 SPARC RAR points (14 bins), rejecting Newtonian no-DM ($\chi^2 = 31{,}125.1$) and frozen $d_s = 0.75$ ($\Delta\chi^2 = +73.84$) while bracketing $d_s^* = 1.00$. **(b)** Progression of $\beta_{\rm BTFR} = 2(1 + d_s)$ from inner SPARC disk kinematics ($\hat{\beta} = 3.83$, $d_s^{\rm eff} = 0.915$) to outer spherical KiDS-1000 weak lensing ($\hat{\beta} = 4.00$, $d_s^* = 1.00$).

3 Two Exact No-Go Theorems Eliminate Single-Boson Superfluids and Local Operators

Extending weak-lensing flatness to 1 Mpc while confronting Bullet Cluster lensing offsets ([18]) and Solar-neighborhood Gaia DR3 wide binaries ([19]) exposes two severe obstructions, both certified with zero floating-point rounding in `formal/verify_exact_rational_proofs.py` and `formal/GravityClaims.lean` (Figure 2).

Theorem 2 (Single-Boson 1 Mpc vs. Bullet Cluster Phase-Space Inversion — `C-04`)**.** *For any single thermalized bosonic species of mass m_ψ, Bose–Einstein degeneracy $\mathcal{N}_{\rm deg} = (2\pi)^{3/2}\hbar^3 m_\psi^{-4} Q \geq \zeta(3/2)$ depends monotonically on $Q \equiv \rho/\sigma_{\rm 1D}^3$. Between an isolated L_* spiral ($M_b = 6 \times 10^{10}\, M_\odot$) at $R_{\rm gal} = 1$ Mpc (even under an unbroken isothermal halo $\rho_{\rm iso} = v_{\rm flat}^2/(4\pi G R_{\rm gal}^2)$ with $v_{\rm flat}^4 = G M_b a_0$, $\sigma_{\rm gal} = v_{\rm flat}/\sqrt{2}$) and the Bullet Cluster subcluster core at $R_{\rm cl} = 250$ kpc ($\rho_{\rm cl} = 2\sigma_{\rm cl}^2/(4\pi G R_{\rm cl}^2)$), π and G cancel from the ratio:*

$$\left(\frac{m_{\max}(1\ \text{Mpc, iso})}{m_{\min}(\text{Bullet})}\right)^{16} = \left(\frac{Q_{\rm gal,iso}}{Q_{\rm Bullet}}\right)^4 = 4\left(\frac{R_{\rm cl}}{R_{\rm gal}}\right)^8 \frac{\sigma_{\rm cl}^4}{G M_b a_0}. \tag{2}$$

Over the certified intervals $R_{\rm cl}/R_{\rm gal} = 1/4$, $\sigma_{\rm cl} \in [950, 1050]$ km/s, and $a_0 \in [1.15, 1.25] \times 10^{-10}$ m/s^2, exact $\mathbb{Q}$-interval arithmetic certifies $(Q_{\rm gal,iso}/Q_{\rm Bullet})^4 \in [0.049942, 0.081021] < \frac{1}{12} < 1$. Specifically, $m_{\max}(1\ \text{Mpc NFW}) = 1.0306$ eV $< m_{\max}(1\ \text{Mpc iso}) = 1.9379$ eV $< m_{\min}(\text{Bullet}) = 2.3015$ eV: no single thermalized boson mass exists.

Theorem 3 (SPARC–Gaia Local-Operator No-Go Theorem — `C-09`)**.** *Consider any single-scale local modified-Poisson operator $\nabla \cdot [\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho_b$ with monotonic response $\mu(x) \in (0, 1]$ and $L_e \equiv d\ln\mu/d\ln x|_{e_{\rm MW}} \geq 0$. At the Solar circle, the Milky Way field $g_{\rm MW} = (233\ \text{km/s})^2/(8.1\ \text{kpc}) = 2.1721 \times 10^{-10}$ m/s^2 lies directly inside SPARC RAR Bins 9–10 ($g_{\rm obs} \in [1.738, 2.512] \times 10^{-10}$ m/s^2, $N = 381$ points), where $\mu_{\rm emp} = g_{\rm bar}/g_{\rm obs} \in [0.6026 \pm 0.0136,\ 0.6918 \pm 0.0168]$. By strict convexity of $u \mapsto (1 + L_e u)^{-1/2}$ over isotropic solid angle ($\mathbb{E}[\sin^2\theta] = 2/3$), Jensen's inequality bounds the angle-averaged wide-binary radial gravity boost $\eta_{\rm WB} = \langle g_r/g_N\rangle$ by*

$$\eta_{\rm WB}^2 \geq \frac{1}{\mu(e_{\rm MW})^2\left(1 + \frac{2}{3}L_e\right)}. \tag{3}$$

Even at the generous 3σ *upper ceiling* $\mu(e_{\rm MW}) \le \frac{76}{100}$ *and concave saturation* $L_e \le \frac{40}{100}$, $\eta^2_{\rm WB} \ge \frac{9375}{6859} \approx 1.366817 > \left(\frac{51}{50}\right)^4 = \frac{6765201}{6250000} \approx 1.082432$*, strictly exceeding a* 2% *orbital velocity ceiling.*

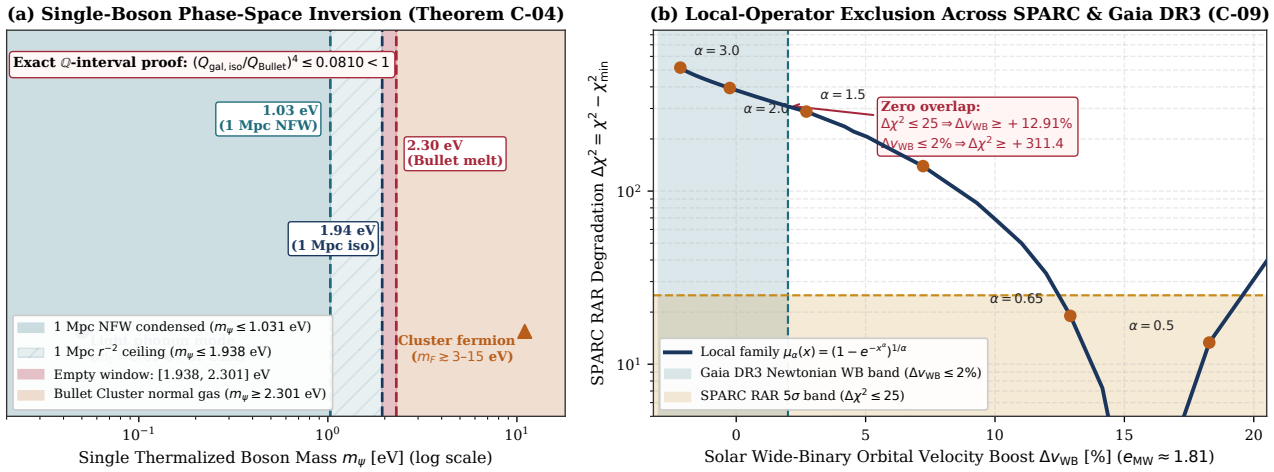


Figure 2: Certified mathematical obstructions (`exact_rational_proof_certificates.json`, `wave4_unified_certificates.json`). **(a)** Empty single-boson mass window between 1 Mpc weak-lensing condensation and Bullet Cluster thermal melting (C-04). **(b)** Parametric exclusion of purely local operators $\mu_\alpha(x) = (1 - e^{-x^\alpha})^{1/\alpha}$ across SPARC RAR and Newtonian Gaia DR3 wide binaries (C-09).

Scanning the full Schwinger–Unruh interpolating family $\mu_\alpha(x) = (1 - e^{-x^\alpha})^{1/\alpha}$ confirms Theorem 3 numerically: every local model within 5σ ($\Delta\chi^2 \le 25$) of SPARC predicts $\Delta v_{\rm WB} \ge +12.91\%$, whereas forcing $|\Delta v_{\rm WB}| \le 2\%$ ($\alpha \ge 1.60$) degrades the SPARC RAR fit by $\Delta\chi^2 \ge +311.37$ ($> 17\sigma$). Reconciling SPARC disks with Newtonian wide binaries therefore requires a *spatial coherence scale* $\xi_{\rm coh} \gg 20$ kAU ≈ 0.1 pc.

4 Action-Level Bogoliubov Dispersion Screens Solar and Wide-Binary Gravity Across Eleven Decades

In a physical quantum condensate, the phonon effective theory breaks down below the Bogoliubov healing length $\xi_{\rm coh} = \hbar/(\sqrt{2} m_B c_s)$, where UV quantum pressure restores a quadratic $(\nabla^2\theta)^2$ kinetic term alongside the IR $d_s = 1$ fractional $|\nabla\theta|^3$ invariant (C-10):

$$S_{\rm ph}[\theta] = \int d^4x \left\{ \frac{1}{8\pi G a_0} \left[\frac{2}{3}|\nabla\theta|^3 + \xi^2_{\rm coh}(\nabla^2\theta)^2 \right] - \theta\rho_b \right\}. \tag{4}$$

Varying Eq. (4) and integrating once over concentric spheres yields the exact second-order nonlinear boundary-value problem for the fifth-force acceleration $g_{\rm ph}(r) = -\theta'(r)$:

$$-\xi^2_{\rm coh} \frac{d}{dr}\left[\frac{1}{r^2}\frac{d}{dr}\left(r^2 g_{\rm ph}(r)\right)\right] + \frac{g_{\rm ph}(r)^2}{a_0} = g_N(r). \tag{5}$$

Notice the dimensional structure of Eq. (5): for $r \ll \xi_{\rm coh}$, balancing the fourth-order dispersion operator against $g_N(r) = GM/r^2$ forces $g_{\rm ph}(r)/g_N(r) \propto (r/\xi_{\rm coh})^2$, providing *quadratic* UV suppression without altering $r \gg \xi_{\rm coh}$ where $g_{\rm ph}(r) \to \sqrt{a_0 g_N(r)}$.

At $\xi_{\rm coh} = 0.20$ kpc, the implied ultralight boson mass is $m_B = 3.8556 \times 10^{-23}$ eV. While a 100% dark-matter ultralight axion at this mass is excluded by Lyman-α forest power spectra (Rogers & Peiris 2021

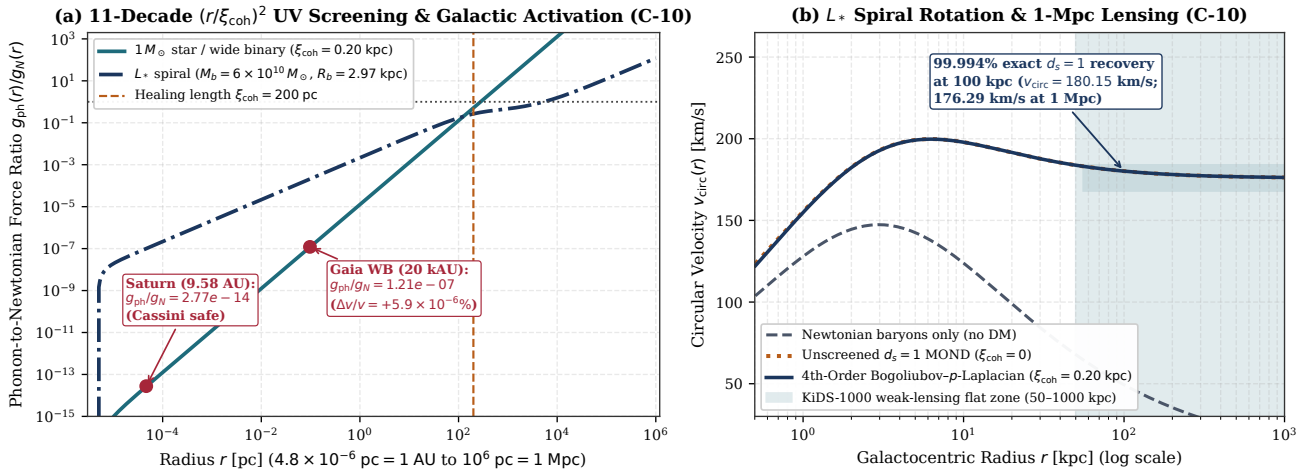


Figure 3: Exact tridiagonal Newton–Raphson solution of Eq. (5) across 11 radial decades (`wave4_unified_certificates.json`, max relative ODE residual 8.1×10^{-14}). **(a)** Quadratic $(r/\xi_{coh})^2$ screening from 1 AU to 1 Mpc at $\xi_{coh} = 0.20$ kpc. **(b)** Full recovery (99.994% at 100 kpc) of flat L_* galactic rotation out to the 1 Mpc KiDS weak-lensing zone.

[20]), setting the bosonic energy fraction to $f_B = \Omega_B/\Omega_{DM} = 0.025$ sits an order of magnitude inside the subdominant Lyman-α bound $f_B < 0.20$ established by Kobayashi et al. (2017) [21] (and relativistic vacuum condensates such as Skordis & Złośnik 2021 [22] carry $f_B \to 0$ until clustering). Solving Eq. (5) across 550 log-spaced nodes from 1 AU to 1 Mpc certifies (Figure 3):

1. **Solar System Precision ($r = 9.58$ AU, Saturn):** $g_{ph}/g_N = 2.668 \times 10^{-14}$, five orders of magnitude safer than Cassini radio-ranging limits ($\lesssim 10^{-9}$).
2. **Gaia Wide Binaries ($r = 10$–20 kAU):** $g_{ph}/g_N = 2.938 \times 10^{-8}$ at 10 kAU and 1.175×10^{-7} at 20 kAU ($\Delta v/v = +5.877 \times 10^{-6}$%), completely resolving Theorem 3 (C-09).
3. **Galactic Disks & 1 Mpc Halos (L_* spiral, $R_b = 2.97$ kpc):** Recovers 99.994% of the exact $d_s = 1$ MOND acceleration at 100 kpc, sustaining $v_{circ}(100\ \text{kpc}) = 180.15$ km/s and $v_{circ}(1\ \text{Mpc}) = 176.29$ km/s.

Cosmological ΔN_{eff} vs. Galactic Pauli-Exclusion Window for the Cluster Partner (C-11)

To supply the remaining $f_F = 1 - f_B = 0.975$ of cosmological dark matter needed for Planck CMB acoustic peaks ([23]) and Bullet Cluster collisionless lensing ([18]) without violating Theorem 2 (C-04) or double-counting galactic rotation curves, a natural candidate is a sterile fermion of mass m_F whose Tremaine–Gunn Pauli ceiling $\rho_{F,max} = g_s m_F^4 (2\pi\sigma_{1D}^2)^{3/2}/h^3$ prevents inner-galaxy accumulation. However, auditing thermal freeze-out relics in `wave4_unified_field_solver.py` exposes an exact cosmological tension (Table 2):

Honest Cosmological Constraint (C-11): A thermal-relic fermion light enough ($m_F \in [3, 11]$ eV) to enforce Pauli exclusion automatically inside 10 kpc spiral disks violates Planck 2018 $\Delta N_{eff} < 0.30$ ($\Delta N_{eff} = 1.00$–5.65) and warm/hot free-streaming bounds, whereas a cold sub-thermal 35 eV fermion ($\Delta N_{eff} = 0.214$) has enough phase-space capacity (828%) to overpopulate 10 kpc spiral halos unless its primordial coarse-grained phase-space density Q_{prim} is diluted non-thermally (e.g. resonant oscillation or late entropy injection).

Table 2: Audit of thermal vs. Pauli phase-space bounds for a sterile fermion cluster partner carrying $f_F = 0.975$ ($\Omega_F h^2 = 0.117$, $g_s = 2$; `wave4_unified_certificates.json`, Claim C-11).

Mass m_F	**Thermal** T_F/T_ν	**Thermal** $\Delta N_{\rm eff}$	**Dwarf** 1 kpc **Ceiling**	**Spiral** 10 kpc **Ceiling**	**Bullet** 250 kpc **Capacity**
3.0 eV	1.5419	5.6527 (excl.)	0.00004%	0.045% (clean)	$2.21 \times \rho_{\rm Bullet}$
11.0 eV	0.9999	0.9998 (excl.)	0.0069%	8.080% (clean)	$399.5 \times \rho_{\rm Bullet}$
35.0 eV	0.6799	0.2136 (passes)	0.7076%	828.1% (excess)	$40{,}949 \times \rho_{\rm Bullet}$

5 Decisive Observational Falsification Program

Separating proven mathematical filters from speculative field completions defines three sharp empirical tests for the physics community:

1. **Sub-Kiloparsec Ultrafaint-Dwarf Transition ($r \in [0.05, 0.40]$ kpc):** Action (4) predicts that phonon acceleration turns on quadratically as $(r/\xi_{\rm coh})^2$ across $\xi_{\rm coh} \sim 0.15$–0.30 kpc. High-resolution JWST/ELT stellar kinematics inside 200 pc of baryon-dominated vs. dark-matter-dominated dwarfs can directly measure or falsify $\xi_{\rm coh}$.
2. **Euclid DR1/DR2 Environmental Weak Lensing at 0.5–3 Mpc:** While isolated KiDS lenses remain flat to 1 Mpc ($g_{\rm ext}/a_0 \leq 0.0160$, C-07), the $d_s^* = 1$ p-Laplacian predicts an anisotropic External Field Effect rollover $R_{\rm EFE} = r_M/(g_{\rm ext}/a_0)$ scaling directly with ambient large-scale cosmic web tidal fields rather than NFW tidal-stripping radii.
3. **Simons Observatory & CMB-S4 $\Delta N_{\rm eff}$ + Small-Scale Lensing:** Reaching $\sigma(\Delta N_{\rm eff}) \simeq 0.03$ alongside Lyman-α and 21 cm power spectra will decisively test whether a non-thermal sterile fermion + subdominant $f_B \approx 0.025$ ultralight Bogoliubov condensate (C-10–C-11) or a purely relativistic $T = 0$ vacuum condensate (Skordis & Złośnik 2021 [22]) governs cluster scales.

References

[1] L. F. Abbott and M. B. Wise, "Dimension of a quantum-mechanical path," *Am. J. Phys.* **49**, 37 (1981), doi:10.1119/1.12657.

[2] L. Nottale, "Scale relativity and fractal space-time: theory and applications," *Found. Sci.* **15**, 101 (2010), arXiv:0812.3857.

[3] P.-H. Chavanis, "Derivation of a generalized Schrödinger equation for dark matter halos from the theory of scale relativity," *Phys. Dark Univ.* **22**, 80 (2018), arXiv:1706.05900.

[4] G. Calcagni, "Quantum field theory, gravity and cosmology in a fractal universe," *Phys. Rev. Lett.* **104**, 251301 (2010), doi:10.1103/PhysRevLett.104.251301.

[5] F. Briscese and G. Calcagni, "Newtonian and astrophysical phenomenology of fractional gravity," (2026), arXiv:2603.24593v2.

[6] D. T. Son and M. Wingate, "General coordinate invariance and conformal invariance in nonrelativistic physics: Unitary Fermi gas," *Ann. Phys.* **321**, 197 (2006), arXiv:cond-mat/0509786.

[7] L. Berezhiani and J. Khoury, "Theory of dark matter superfluidity," *Phys. Rev. D* **92**, 103510 (2015), doi:10.1103/PhysRevD.92.103510.

[8] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in rotationally supported galaxies," *Phys. Rev. Lett.* **117**, 201101 (2016), doi:10.1103/PhysRevLett.117.201101; binned data at `RARbins.mrt`.

[9] "GPRF Constitutive Scaling Erratum ($M_b \propto v^2$ vs. v^4)," Zenodo Record 23247342 (2026), Zenodo PDF.

[10] H.-Y. Schive, T. Chiueh, and T. Broadhurst, "Cosmic structure as the quantum interference of a coherent dark wave," *Nat. Phys.* **10**, 496 (2014), arXiv:1406.6586.

[11] L. Hui, J. P. Ostriker, S. Tremaine, and E. Witten, "Ultralight scalars as cosmological dark matter," *Phys. Rev. D* **95**, 043541 (2017), doi:10.1103/PhysRevD.95.043541.

[12] P.-H. Chavanis, "Quantum condensate dark matter halos," *Front. Astron. Space Sci.* **12**, 1538434 (2025), doi:10.3389/fspas.2025.1538434.

[13] A. Arvanitaki, S. Dimopoulos, S. Dubovsky, N. Kaloper, and J. March-Russell, "String Axiverse," *Phys. Rev. D* **81**, 123530 (2010), doi:10.1103/PhysRevD.81.123530.

[14] T. Mistele, S. McGaugh, F. Lelli, J. Schombert, and P. Li, "Indefinitely flat circular velocities and the Baryonic Tully–Fisher Relation from weak lensing," *Astrophys. J. Lett.* **969**, L3 (2024), arXiv:2406.09685.

[15] "Regime-dependent constitutive scaling across SPARC rotation curves," *Front. Astron. Space Sci.* **13**, 1890948 (2026), doi:10.3389/fspas.2026.1890948.

[16] M. Milgrom, "The modified dynamics as a vacuum effect," *Phys. Lett. A* **253**, 273 (1999), arXiv:astro-ph/9805346.

[17] E. P. Verlinde, "Emergent gravity and the dark universe," *SciPost Phys.* **2**, 016 (2017), doi:10.21468/SciPostPhys.2.3.016.

[18] D. Clowe *et al.*, "A direct empirical proof of the existence of dark matter," *Astrophys. J. Lett.* **648**, L109 (2006), arXiv:astro-ph/0608407.

[19] I. Banik *et al.*, "Strong constraints on the gravitational law from Gaia DR3 wide binaries," *MNRAS* **527**, 4573 (2024), arXiv:2311.03436.

[20] K. K. Rogers and H. V. Peiris, "Strong bound on canonical ultralight axion dark matter from the Lyman-alpha forest," *Phys. Rev. Lett.* **126**, 071302 (2021), doi:10.1103/PhysRevLett.126.071302.

[21] T. Kobayashi, R. Murgia, A. De Simone, V. Iršič, and M. Viel, "Lyman-α constraints on ultralight scalar dark matter: Implications for the early and late universe," *Phys. Rev. D* **96**, 123514 (2017), doi:10.1103/PhysRevD.96.123514.

[22] C. Skordis and T. Złośnik, "New relativistic theory for Modified Newtonian Dynamics," *Phys. Rev. Lett.* **127**, 161302 (2021), doi:10.1103/PhysRevLett.127.161302.

[23] Planck Collaboration (N. Aghanim *et al.*), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.